

Chemistry Practice Quizzes — Bonding, Ions & Formula Writing

Two parallel practice quizzes for the upcoming chemistry quiz test. Each has 17 questions spread across the same seven topics in the same proportions — use Quiz A first, then Quiz B a day or two later as a second, independent check (the questions are different, so it isn't just recall of Quiz A's answers).

Question distribution (same in both quizzes):

Topic	Questions
Electronegativity & bond character	3
Metallic bonds	2
Ionic bonds	2
Covalent bonds	2
Polyatomic ions (ratios & writing)	3
Writing formulas with prefixes (covalent + ionic)	3
Alloys	2

How to use this: attempt a full quiz closed-book, in one sitting, before checking the answer key at the end. Treat it like the real thing — no notes, no flashcards mid-quiz.

Quiz A

- Which pair of atoms would form a purely *nonpolar* covalent bond? a) Na and Cl b) H and F c) Cl and Cl d) Mg and O
- Using electronegativities Na = 0.9 and Cl = 3.0, calculate ΔEN for a Na-Cl bond and state whether it's best described as nonpolar covalent, polar covalent, or ionic.
- Fill in the blanks: in a polar covalent bond, the shared electron pair is pulled toward the atom with the _____ electronegativity, giving that atom a partial _____ charge (symbol: _____).
- Which two properties are explained by metal-ion layers sliding past each other while the surrounding electron sea keeps the bond intact? a) Malleability and ductility b) Lustre and sonority c) High melting point and density d) Conductivity and reactivity
- Define a metallic bond in one sentence.
- An ionic bond forms as a direct result of: a) two atoms sharing an electron pair equally b) one atom completely transferring an electron to another atom, then the resulting ions attracting each other c) two identical atoms bonding together d) a metal atom bonding with another metal atom
- Fill in the blanks: when a sodium atom loses its one valence electron, it becomes a _____ (cation/anion) called Na^+ ; when a chlorine atom gains one electron, it becomes a(n) _____ (cation/anion) called Cl^- .
- What is the key difference between how electrons behave in a covalent bond versus an ionic bond?
- Which of these is most likely to be a covalent compound rather than an ionic one? a) NaCl (Na EN 0.9, Cl EN 3.0) b) MgO (Mg EN 1.2, O EN 3.5) c) CO_2 (C EN 2.5, O EN 3.5) d) $CaCl_2$ (Ca EN 1.0, Cl EN 3.0)
- What is a polyatomic ion? Give one example with its formula and charge.
- Write the formula for calcium nitrate (Ca^{2+} and NO_3^-).

- 12.** Brackets are required around a polyatomic ion in a chemical formula when: a) the ion has a negative charge b) two or more of that same polyatomic ion are needed in the formula c) the compound is covalent d) never — brackets are always optional
- 13.** Name the compound N_2O_5 using the Greek prefix system.
- 14.** Write the chemical formula for “sulfur hexafluoride.”
- 15.** Explain why H_2S is called “hydrogen sulfide,” not “dihydrogen sulfide.”
- 16.** Which of the following is an accurate description of an alloy? a) A pure element with unusually strong bonds b) A chemical compound formed by a reaction between two metals c) A homogeneous mixture of two or more metals, or a metal and a non-metal d) Any metal found in nature
- 17.** Name one alloy, state its two components, and explain briefly why it counts as a mixture rather than a compound.
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Quiz B

- 1.** Which pair would you expect to form the *most ionic* bond (largest ΔEN)? a) H and Cl (EN 2.1, 3.0) b) C and H (EN 2.5, 2.1) c) Na and F (EN 0.9, 4.0) d) O and O (EN 3.5, 3.5)
- 2.** Electronegativities: Mg = 1.2, O = 3.5. Calculate ΔEN for Mg–O and classify the bond.
- 3.** Fill in the blanks: two identical atoms bonding together (like O_2 or F_2) always form a _____ covalent bond because $\Delta\text{EN} = \text{_____}$.
- 4.** Which property of pure metals is explained by the free electrons in the electron sea reflecting light? a) Ductility b) Lustre c) Sonority d) High melting point
- 5.** List three physical properties of pure metals that follow from the electron-sea model (other than lustre).
- 6.** In the formation of MgCl_2 from Mg and Cl atoms, magnesium: a) shares two electrons with each chlorine atom b) gains two electrons to become Mg^{2-} c) loses two electrons (one to each of two chlorine atoms) to become Mg^{2+} d) does not change its charge
- 7.** Fill in the blank: the electrostatic force of attraction between oppositely charged ions is called a(n) _____ bond.
- 8.** Would you expect a covalent compound like sugar to conduct electricity when dissolved in water? Explain briefly.
- 9.** Which best describes what happens to the shared electron pair in a polar covalent bond like H–F? a) It sits exactly between the two nuclei b) It transfers completely to fluorine c) It is displaced toward the more electronegative atom (F) but still shared d) It transfers completely to hydrogen
- 10.** What is the difference between a monoatomic ion and a polyatomic ion?
- 11.** Write the formula for aluminium hydroxide (Al^{3+} and OH^-).
- 12.** In the formula $(\text{NH}_4)_2\text{SO}_4$, why are there brackets around NH_4 ? a) Ammonium is a metal b) Because two ammonium ions are needed to balance one sulfate ion’s charge c) Because ammonium is negative d) Brackets are never required for cations
- 13.** Name the compound PCl_3 using the Greek prefix system.
- 14.** Write the chemical formula for “carbon disulfide.”
- 15.** Why is CO_2 named “carbon dioxide” and not “monocarbon dioxide”?
- 16.** Which statement about alloys is TRUE? a) Alloys always conduct electricity better than the pure metals they’re made from b) Alloys generally conduct electricity and heat less well, and melt at a lower temperature, than the pure metals they’re made from c) An alloy is always a compound of a metal and oxygen d) Alloys cannot contain non-metals

17. Stainless steel is described as iron mixed with nickel and chromium. Explain in one sentence why alloying iron this way is useful.

Answer Key

Quiz A

1. **c) Cl and Cl** — identical atoms, $\Delta EN = 0$, so the shared pair sits exactly between the two nuclei with neither atom pulling harder.
2. $\Delta EN = |0.9 - 3.0| = 2.1 \rightarrow$ **ionic** (well past the ~ 1.7 rule-of-thumb line).
3. **higher; negative; δ^- .**
4. **a) Malleability and ductility.**
5. The electrostatic attraction between a lattice of positive metal ions and the surrounding sea of delocalized (mobile) electrons.
6. **b)** — complete electron transfer, then electrostatic attraction between the resulting ions.
7. **cation; anion.**
8. In a covalent bond the electron pair is *shared* between the two atoms; in an ionic bond one or more electrons are *completely transferred* from one atom to the other.
9. **c) CO_2** — $\Delta EN = 1.0$, much smaller than the other three pairs, and both atoms are non-metals.
10. An ion made of two or more atoms covalently bonded together that carries an overall charge — e.g. sulfate, SO_4^{2-} .
11. **$\text{Ca}(\text{NO}_3)_2$** — two nitrate ions are needed to balance calcium's 2+ charge, so brackets are required.
12. **b)** — two or more of the same polyatomic ion are needed in the formula.
13. **Dinitrogen pentoxide.**
14. **SF_6 .**
15. Hydrogen never takes a prefix when it is the first element in the formula, no matter how many atoms are present.
16. **c)** — a homogeneous mixture of two or more metals, or a metal and a non-metal.
17. E.g. brass = copper + zinc. It's a mixture, not a compound, because the metals are physically combined (melted together) without forming a new chemical bond between the copper and zinc atoms — no fixed ratio of chemical bonding, just a blended solid.

Quiz B

1. **c) Na and F** — $\Delta EN = 3.1$, the largest of the four pairs.
2. $\Delta EN = |1.2 - 3.5| = 2.3 \rightarrow$ **ionic.**
3. **nonpolar; 0.**
4. **b) Lustre.**
5. Any three of: malleable, ductile, good conductor of heat/electricity, sonorous, generally high melting point.
6. **c)** — Mg loses two electrons total (one to each Cl atom) to become Mg^{2+} .
7. **ionic.**
8. No — covalent compounds generally don't release ions when dissolved, so there are no free charge carriers to conduct electricity; this is the opposite of ionic compounds, which do conduct once dissolved because they release mobile ions.
9. **c)** — displaced toward the more electronegative atom but still shared, giving a dipole moment.
10. A monoatomic ion forms from a single atom that has gained or lost electrons; a polyatomic ion forms from two or more atoms bonded together that together carry one overall charge.
11. **$\text{Al}(\text{OH})_3$.**
12. **b)** — two ammonium ions are needed to balance one sulfate ion's 2- charge.
13. **Phosphorus trichloride.**
14. **CS_2 .**

15. The prefix *mono-* is always dropped for the first element in a formula — carbon needs no prefix here.
16. **b).**
17. Mixing in nickel and chromium makes the iron harder, stronger, and resistant to rust (corrosion) — a clear improvement over pure iron, which is soft and stretches easily when hot.